

Developing a Basic GUI Application Using JavaFX in Eclipse

This tutorial takes readers through the process of developing a basic GUI application using JavaFX in Eclipse, and is simple enough for even a beginner to follow.



A graphical user interface or GUI is a computer program that makes it easy to talk to your device. GUIs are used in almost every electronic device worldwide owing to their ease of use, interactive designs and higher accessibility. Even the OS installed in your computer is an example of a GUI. In this tutorial, we will look at developing a basic GUI in Java using the JavaFX library. The only pre-requisite for this tutorial is that you need a basic understanding of the Java language and of the Eclipse IDE. But before that, let's take an overview of JavaFX.

JavaFX: The future

Oracle defines JavaFX as "...the next step in the evolution of Java as a rich client platform." JavaFX was first launched by Sun Microsystems in 2007 and JavaFX 1.0 was released in December 2008. The latest release is JavaFX 8 and it comes bundled with JRE/JDK for Java 8; hence, the name. It

is widely regarded as the future in developing sophisticated applications – whether Web based, desktop based or for the mobile – and is rapidly replacing Swing owing to its various advantages. These include being lightweight, having CSS styling, sleek design controls, and the use of FXML and Scene Builder.

Scene Builder and its role in GUI development

Oracle.com defines Scene Builder as "...a visual layout tool that lets users quickly design JavaFX application user interfaces, without coding." That's right—you can actually design different controls in your application, like textboxes, dropdowns, labels, buttons, etc, without actually coding them in your favourite integrated development environment (IDE). Isn't that easy and time saving? This is a standalone tool, which can be embedded in your IDE workspace for rapid development. A drag-and-drop feature is available (the same